

GIA Training School (Gävle 2023): References for Lectures by Glenn Milne

- Afonso, J.C., Salajegheh, F., Szwillus, W., Ebbing, J., Gaina, C., 2019. A global reference model of the lithosphere and upper mantle from joint inversion and analysis of multiple data sets. *Geophys. J. Int.* 217, 1602e1628.
- Clark, P. U. and Tarasov, L.: Closing the sea level budget at the Last Glacial Maximum, *P. Natl. Acad. Sci. USA*, 111, 15861–15862, <https://doi.org/10.1073/pnas.1418970111>, 2014.
- Dalca, A. V., Ferrier, K. L., Mitrovica, J. X., Perron, J. T., Milne, G. A., and Creveling, J. R.: On postglacial sea level-III. Incorporating sediment redistribution, *Geophys. J. Int.*, 194, 45–60, <https://doi.org/10.1093/gji/ggt089>, 2013.
- Dalton, A.S. et al.: An updated radiocarbon-based ice margin chronology for the last deglaciation of the North American Ice Sheet Complex. *Quat. Sci. Rev.* 234, 106223. <https://doi.org/10.1016/j.quascirev.2020.106223>. ISSN 0277-3791.
- Dutton, A., Carlson, A. E., Long, A. J., Milne, G. A., Clark, P. U., DeConto, R., Horton, B. P., Rahmstorf, S., and Raymo, M. E.: Sea-level rise due to polar ice-sheet mass loss during past warm periods, *Science*, 349, aaa4019, <https://doi.org/10.1126/science.aaa4019>, 2015.
- Farrell, W. E. and J. A. Clark (1976). "On Postglacial Sea level." *Geophysical Journal of the Royal Astronomical Society* **46**: 647-667.
- Ferrier, K. L., Mitrovica, J. X., Giosan, L., and Clift, P. D.: Sea-level responses to erosion and deposition of sediment in the Indus River basin and the Arabian Sea, *Earth Planet. Sc. Lett.*, 416, 12–20, <https://doi.org/10.1016/j.epsl.2015.01.026>, 2015.
- Hughes, A. L. C., Gyllencreutz, R., Lohne, O. S., Mangerud, J., and Svendsen, J. I.: The last Eurasian ice sheets – a chronological and time-slice reconstruction, *DATED-1, Boreas*, 45, 1–45, <https://doi.org/10.1111/bor.12142>, 2016.
- Ivins, E. R., James, T. S., Wahr, J., Schrama, E. J. O., Landerer, F. W., and Simon, K. M.: Antarctic Contribution to Sea-Level Rise Observed by GRACE with Improved GIA Correction, *J. Geophys. Res.-Sol. Ea.*, 118, 3126–3141, <https://doi.org/10.1002/jgrb.50208>, 2013.
- Knipe, R.J., "Deformation mechanisms - recognition from natural tectonites", *Journal of Structural Geology*, 11, 127-146, 1989.
- Kuchar, J., Milne, G., Wolstencroft, M., Love, R., Tarasov, L., & Hijma, M. (2018). The influence of sediment isostatic adjustment on sea level change and land motion along the U.S. Gulf Coast. *Journal of Geophysical Research: Solid Earth*, 123, 780–796. <https://doi.org/10.1002/2017JB014695>Kuchar et al (2020)
- Lambeck, K., Rouby, H., Purcell, A., Sun, Y. Y., and Sambridge, M.: Sea level and global ice volumes from the Last Glacial Maximum to the Holocene, *P. Natl. Acad. Sci. USA*, 111, 15296–15303, <https://doi.org/10.1073/pnas.1411762111>, 2014.
- Lambeck, K., C. Smither, et al. (1998). "Sea-level change, glacial rebound and mantle viscosity for northern Europe." *Geophysical Journal International*, **134**: 102-144.
- Lisiecki, L. E., and M. E. Raymo (2005), A Pliocene-Pleistocene stack of 57 globally distributed benthic $d^{18}O$ records, *Paleoceanography*, 20, PA1003, doi:10.1029/2004PA001071.
- Lecavalier, B. S., Milne, G. A., Simpson, M. J. R., Wake, L., Huybrechts, P., Tarasov, L., Kjeldsen, K. K., Funder, S., Long, A. J., Woodroffe, S., Dyke, A. S., and Larsen, N. K.: A model of Greenland ice sheet deglaciation constrained by observations of relative sea level and ice extent, *Quaternary Sci. Rev.*, 102, 54–84, <https://doi.org/10.1016/j.quascirev.2014.07.018>, 2014.

- Love, R., G. A. Milne, L. Tarasov, S. E. Engelhart, M. P. Hijma, K. Latychev, B. P. Horton, and T. E. Törnqvist (2016), The contribution of glacial isostatic adjustment to projections of sea-level change along the Atlantic and Gulf coasts of North America, *Earth's Future*, 4, 440–464, doi:10.1002/2016EF000363.
- Means, W.D., “Stress and Strain: Basic Concepts of Continuum Mechanics for Geologists”, Springer-Verlag, New York, 1976.
- Milne, G.A. and I. Shennan, “Modelling Glaciation-Induced Sea-Level Changes” in *Encyclopedia of Quaternary Sciences*, volume on Sea-Level Change, edited by Ian Shennan, Elsevier Publications, London, 2013.
- Milne, G.A. and J.X. Mitrovica, “Searching for Eustasy in Deglacial Sea-Level Histories”, *Quat. Sci. Rev.*, 27, 2292-2302, 2008.
- Milne, G.A., Latychev, K., Schaeffer, A., Crowley, J.W., Lecavalier, B.S., Audette, A., 2018. The influence of lateral Earth structure on glacial isostatic adjustment in Greenland. *Geophys. J. Int.* 214, 1252–1266. <https://doi.org/10.1093/gji/ggy189>.
- Mitrovica, J.X., “Haskell [1935] Revisited”, *J. Geophys. Res.*, 101, 555-569, 1996.
- Mitrovica, J. X. and Wahr, J.: Ice Age Earth Rotation, *Annu. Rev. Earth Pl. Sc.*, 39, 577–616, <https://doi.org/10.1146/annurevearth-040610-133404>, 2011.
- Mitrovica, J.X. and A.M. Forte, "A New Inference of Mantle Viscosity Based Upon a Joint Inversion of Convection and Glacial Isostatic Adjustment Data", *Earth Planet. Sci. Lett.*, 225, 177-189, 2004.
- Mitrovica, J.X. and G.A. Milne, “On the Origin of Postglacial Ocean Syphoning”, *Quat. Sci. Rev.*, 21, 2179-2190, 2002.
- Mitrovica, J.X. & Peltier, W.R., 1993. A new formalism for inferring mantle viscosity based on estimates of post glacial decay times: application to RSL variations in N.E. Hudson Bay, *Geophys. Res. Lett.*, 20, 2183–2186.
- Mitrovica JX, Milne GA, Davis JL. 2001. Glacial isostatic adjustment on a rotating Earth. *Geophys. J. Int.* 147:562–78
- Mitrovica, J.X., M.E. Tamiseia, J.L. Davis, and G.A. Milne, “Recent Mass Balance of Polar Ice Sheets Inferred from Patterns of Global Sea-Level Change”, *Nature*, 409, 102-1029, 2001b
- Moucha, R. and Ruetenik, G. A.: Interplay between dynamic topography and flexure along the US Atlantic passive margin: Insights from landscape evolution, *Global Planet. Change*, 149, 72–78, <https://doi.org/10.1016/j.gloplacha.2017.01.004>, 2017.
- Peltier, W. R. (1974). "The impulse response of a Maxwell earth." *Rev. geophys. space Phys.* 12: 649.
- Peltier, W. R.: Global glacial isostasy and the surface of the ice-age earth: The ICE-5G (VM2) model and GRACE, *Annu. Rev. Earth Pl. Sc.*, 32, 111–149, 2004.
- Ranalli, G., “Rheology of the Earth”, Chapman and Hall, London, 1995.
- Ritsema, J., Deuss, A., van Heijst, H.J., Woodhouse, J.H., 2011. S40RTS: a degree-40 shear-velocity model for the mantle from new Rayleigh wave dispersion, teleseismic traveltime and normal-mode splitting function measurements. *Geophys. J. Int.* 184, 1223e1236.
- Ruddiman, W. E., “Earth’s Climate: Past and Future”, W. H. Freeman, New York, 2013.
- Simms, A.R., Lisiecki, L., Gebbie, G., Whitehouse, P.L., Clark, J.F., 2019. Balancing the last glacial maximum (LGM) sea-level budget. *Quat. Sci. Rev.* 205, 143e153.
- Steffen et al. (2020)
- Tarasov, L., Dyke, A. S., Neal, R. M., and Peltier, W. R.: A data-calibrated distribution of deglacial chronologies for the North American ice complex from glaciological modeling, *Earth Planet. Sc. Lett.*, 315, 30–40, <https://doi.org/10.1016/j.epsl.2011.09.010>, 2012.
- Whitehouse, P. L., Bentley, M. J., Milne, G. A., King, M. A., and Thomas, I. D.: A new glacial isostatic adjustment model for Antarctica: calibrating the deglacial model

- using observations of relative sea-level and present-day uplift rates, *Geophys. J. Int.*, 190, 1464–1482, <https://doi.org/10.1111/j.1365-246X.2012.05557.x>, 2012.
- Whitehouse, P. L.: Glacial isostatic adjustment modelling: historical perspectives, recent advances, and future directions, *Earth Surf. Dynam.*, 6, 401–429, 2018
<https://doi.org/10.5194/esurf-6-401-2018>
- Wilson, R.C.L., S.A. Drury and J.L. Chapman, “The Great Ice Age: Climate Change and Life”, Routledge, London, 2000.
- Wu, P. and W. R. Peltier (1982). "Viscous gravitational relaxation." *Geophysical Journal of the Royal Astronomical Society* **70**: 435-485.